

PATTANI, Thailand

WMO: 485800

Lat: 6.7897N

Lon: 101.1469E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.5	22.3	19.3	14.0	26.7	20.0	14.7	26.6	5.4	29.1	5.0	29.2	0.1	120	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.4	35.0	26.4	34.3	26.3	33.7	26.2	27.7	32.8	27.2	32.3	27.0	32.1	2.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	30.4	26.0	21.3	30.0	25.7	20.9	29.6	88.1	33.1	86.4	32.6	85.1	32.3	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.4	3.8	DB	20.1	36.0	1.5	1.0	19.1	36.7	18.2	37.4	17.4	37.9	16.3	38.7	(4)
(5)			WB	19.1	28.6	1.8	1.0	17.8	29.3	16.7	29.9	15.7	30.5	14.4	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	26.7	27.3	28.2	29.1	29.3	28.9	28.5	28.5	28.3	27.8	27.2	26.7	(6)	
	DBStd	1.27	0.99	0.96	0.96	0.91	0.92	0.80	0.84	0.86	0.75	0.88	1.14	1.16	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6585	519	484	564	573	597	567	575	573	548	552	515	517	(10)	
	CDD18.3	3543	261	251	306	323	338	317	317	315	298	294	265	258	(11)	
	CDH23.3	35884	2262	2405	3232	3722	3828	3494	3380	3306	2978	2767	2366	2145	(12)	
	CDH26.7	14114	744	969	1429	1719	1687	1484	1393	1357	1119	928	695	590	(13)	
(14) Wind	WSAvg	0.8	0.9	1.0	0.9	0.8	0.7	0.7	0.8	0.9	0.8	0.7	0.5	0.7	(14)	
(15) Precipitation	PrecAvg	1919	75	37	63	79	129	111	119	144	138	232	394	397	(15)	
	PrecMax	2592	331	280	274	242	298	257	211	322	213	474	886	967	(16)	
	PrecMin	1358	2	0	0	0	28	15	43	74	52	98	153	37	(17)	
	PrecStd	371	83	63	68	59	57	52	45	61	39	75	182	237	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.9	35.1	36.0	35.8	34.8	34.5	34.5	33.8	33.4	32.6	32.4	(19)	
		MCWB	25.5	25.4	26.0	26.9	26.5	26.4	25.9	25.8	26.1	26.3	26.2	25.9	(20)	
	2%	DB	31.8	32.9	34.1	35.0	34.7	34.0	33.6	33.7	33.1	32.6	31.9	31.5	(21)	
		MCWB	25.3	25.3	26.0	26.6	26.6	26.4	26.0	25.9	26.2	26.2	26.1	25.7	(22)	
	5%	DB	31.1	32.2	33.4	34.2	33.9	33.4	33.0	33.0	32.4	31.8	31.2	30.8	(23)	
		MCWB	25.0	25.2	25.8	26.4	26.6	26.3	26.0	26.0	26.1	26.0	26.0	25.5	(24)	
	10%	DB	30.3	31.5	32.6	33.3	33.1	32.6	32.3	32.2	31.7	31.0	30.3	29.9	(25)	
		MCWB	24.8	24.9	25.7	26.3	26.5	26.2	25.9	25.8	25.9	25.9	25.9	25.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.9	27.7	28.2	28.0	27.9	27.5	27.5	27.7	27.2	27.2	26.9	(27)	
		MCDB	31.1	32.2	33.3	34.1	33.6	32.8	32.3	32.3	32.0	31.5	31.1	30.7	(28)	
	2%	WB	26.1	26.3	27.0	27.4	27.5	27.2	26.9	26.9	27.0	26.7	26.7	26.3	(29)	
		MCDB	30.4	31.4	32.5	33.2	32.9	32.1	31.8	31.9	31.4	30.9	30.5	30.1	(30)	
	5%	WB	25.6	25.8	26.5	27.0	27.1	26.9	26.5	26.5	26.5	26.4	26.4	25.9	(31)	
		MCDB	29.7	30.8	31.9	32.8	32.4	31.7	31.4	31.4	31.0	30.5	30.0	29.5	(32)	
	10%	WB	25.2	25.4	26.1	26.6	26.7	26.5	26.2	26.1	26.1	26.1	26.0	25.5	(33)	
		MCDB	29.0	30.1	31.3	32.0	31.7	31.3	30.9	30.9	30.6	30.1	29.5	28.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.9	9.3	9.1	8.4	8.2	8.1	8.2	7.6	6.9	5.9	5.9	(35)	
		MCDBR	8.6	9.8	10.4	10.1	9.3	8.9	8.9	9.0	8.5	8.1	7.5	7.6	(36)	
	5% WB	MCWBR	3.0	3.4	3.5	3.3	3.0	3.0	3.0	3.1	2.9	3.0	2.8	2.9	(37)	
		MCWBR	7.4	8.8	9.4	9.3	8.7	8.4	8.5	8.6	8.0	7.5	6.8	6.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.461	0.461	0.470	0.457	0.450	0.461	0.465	0.463	0.454	0.453	0.453	0.465	(40)	
		taud	2.297	2.272	2.270	2.310	2.353	2.350	2.329	2.321	2.340	2.344	2.349	2.299	(41)	
	Ebn at Noon		857	867	859	856	844	824	824	840	860	863	859	845	(42)	
		Edh at Noon	134	140	142	134	125	124	127	131	130	129	126	132	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.70	5.70	5.87	5.89	5.26	5.03	4.98	5.17	5.05	4.63	4.08	3.93	(44)	
	RadStd		0.35	0.37	0.48	0.26	0.29	0.20	0.24	0.16	0.18	0.31	0.35	0.43	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.64	+0.27	+0.31	N/A	N/A	N/A	N/A	(46)
(47)	Regional (13 neighbors)	+0.20	N/A	N/A	+0.30	+0.25	+0.19	N/A	N/A	+78	+79	(47)

Nomenclature: See separate page